

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 06



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Recap of Previous Lecture



Topic

Complement of a graph



Topic

Graph isomorphism



Topics to be Covered



Topic

Graph isomorphism



Topic

Self-complementary graph

Topic

Planar graphs

Topic

Region and degree of regions

Topic

Sum of degree theorem for regions





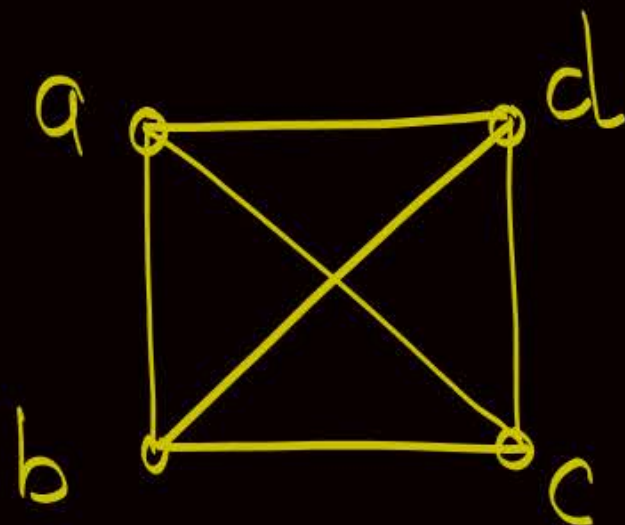
Topic : Graph Isomorphism

Two graphs are said to be isomorphic if their properties are identical (name and look need not be same)

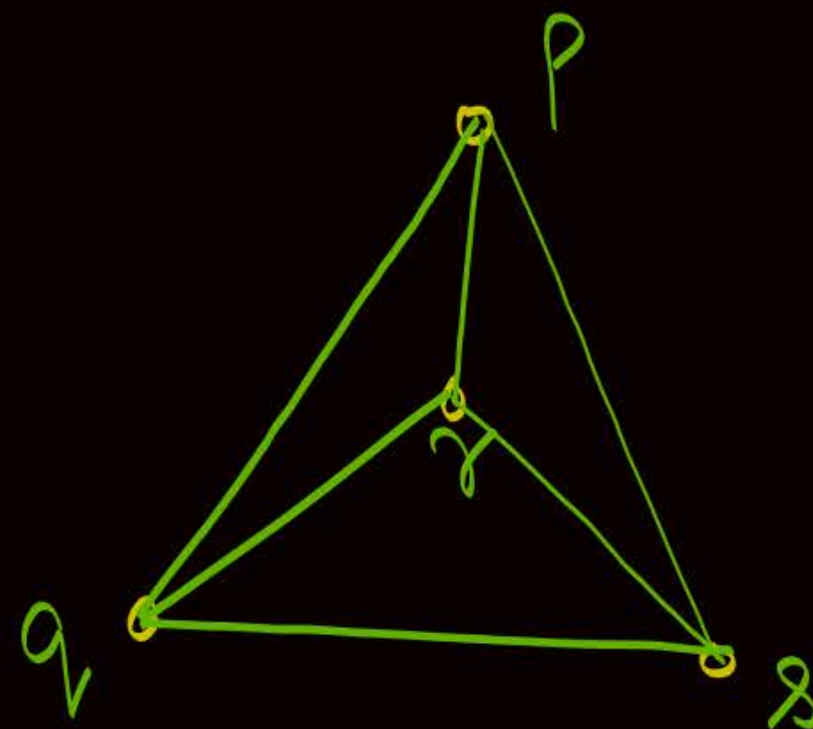
Two graphs G and G' are said to be isomorphic if there exists a function $f: V(G) \rightarrow V(G')$ such that

- I. f is bijective i.e., $|V(G)| = |V(G')|$ and for every vertex of set $V(G)$ there is a unique image in $V(G')$
- II. f preserves adjacency of vertices

└ i.e., two vertices of graph G are adjacent if and only if their corresponding images in G' are adjacent } $\therefore |E(G)| = |E(G')|$



G_1



G_2

$$V(G_1) = \{a, b, c, d\}$$

$$V(G_2) = (p, q, r, s)$$

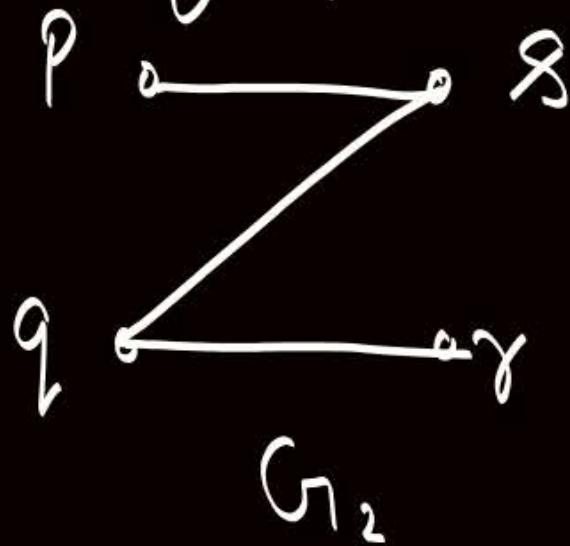
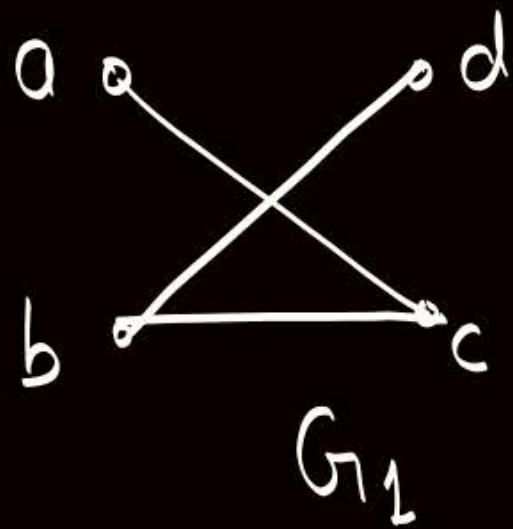
$$\left. \begin{array}{l} f(a) = p \\ f(b) = q \\ f(c) = r \\ f(d) = s \end{array} \right\}$$

This bijective function
exists and it preserves
the adjacency of vertices
 $\therefore G_1$ and G_2 are isomorphic

G is isomorphic to G' is
denoted by

$$G \cong G'$$

Q: Consider two graphs G_1 and G_2 .
Check whether the graphs are isomorphic or not



$|V(G_1)| = |V(G_2)|$
 and $|E(G_1)| = |E(G_2)|$

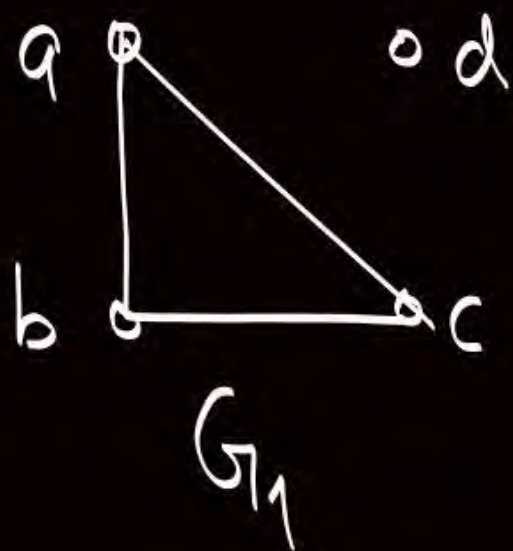
$\left. \begin{array}{l} |V(G_1)| = |V(G_2)| \\ \text{and } |E(G_1)| = |E(G_2)| \end{array} \right\} \begin{array}{l} \therefore \text{Graphs} \\ \text{may be} \\ \text{isomorphic} \\ \text{(not necessarily)} \end{array}$

$$\begin{aligned} f(a) &= p \\ f(b) &= q \\ f(c) &= r \\ f(d) &= s \end{aligned}$$

This bijection is possible,
 s.t. it preserves the
 adjacency of vertices

$\therefore G_1$ & G_2
 are isomorphic

Q: Consider two graphs G_1 and G_2 .
Check whether the graphs are isomorphic or not



$$\left. \begin{array}{l} |V(G_1)| = |V(G_2)| \\ \text{and } |E(G_1)| = |E(G_2)| \end{array} \right\}$$

Condition holds true,
but graphs
are not isomorphic



Topic : Graph Isomorphism

If $G \cong G'$, then following conditions must hold true.

These four conditions are necessary conditions for two graphs to be isomorphic. If any of the four conditions is dissatisfied, then graphs cannot be isomorphic.

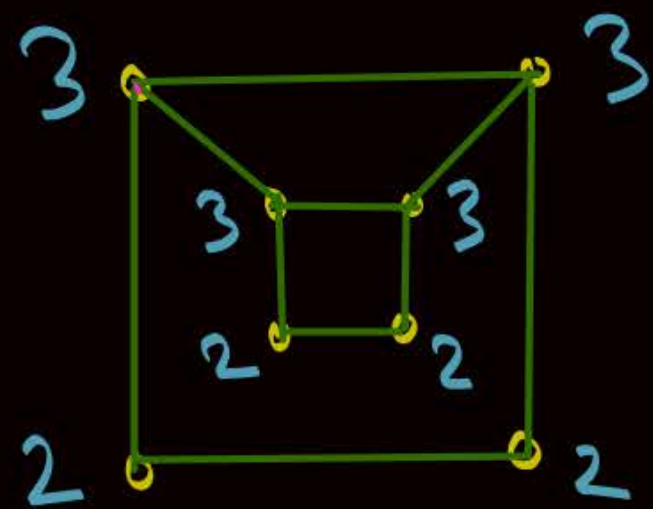
① $|V(G)| = |V(G')|$

② $|E(G)| = |E(G')|$

③ Degree sequence of $G =$ Degree sequence of G'

④ If some 'k' vertices $v_1, v_2, \dots, v_k$ in graph G form a cycle of length 'k', then their corresponding images in G' i.e. $f(v_1), f(v_2), f(v_3), \dots, f(v_k)$ must also form a cycle of length 'k'.

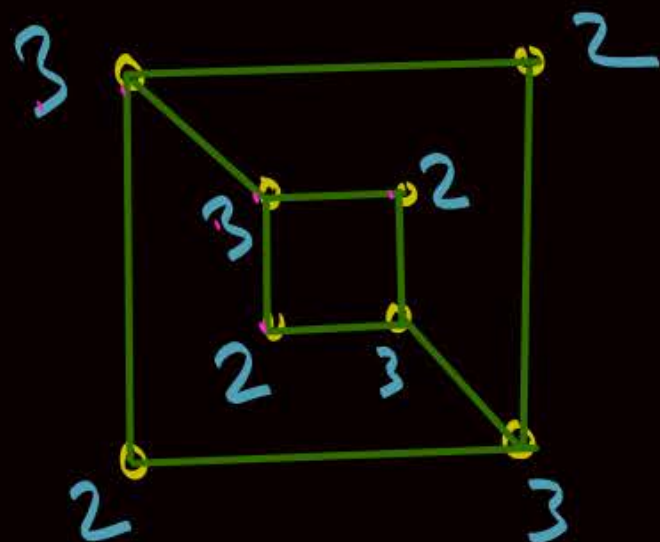
H.W. Q: Consider two graphs G_1 & G_2



G_1

Deg seq. of $G_1 = \{3, 3, 3, 3, 2, 2, 2, 2\}$

Check whether the graphs are isomorphic or not?



G_2

Deg seq. of $G_2 = \{3, 3, 3, 3, 2, 2, 2, 2\}$

✓ ① $|V(G_1)| = |V(G_2)|$

✓ ② $|E(G_1)| = |E(G_2)|$

✓ ③ Deg. seq. of $G_1 = \text{Deg. seq. of } G_2$

✗ ④ The only 4 vertices of deg=3 in G_1 form a cycle of length '4', but vertices of deg=3 in G_2 does not form any cycle

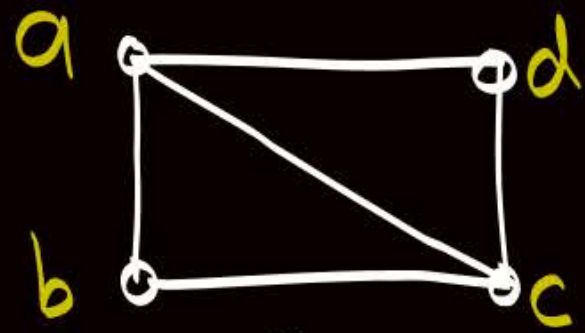
∴ G_1 & G_2 are not isomorphic



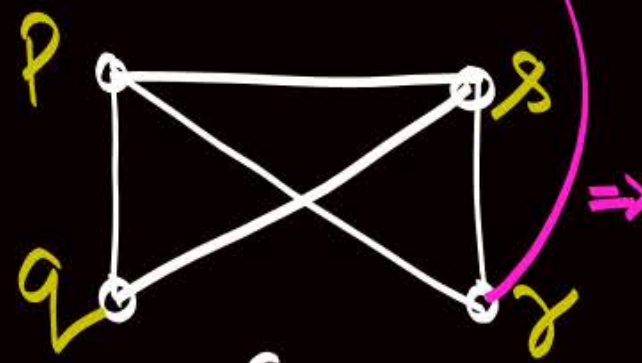
Topic : Graph Isomorphism

→ Note:- Two simple graphs G_1 and G_2 are isomorphic to each other if and only if their complements are isomorphic to each other

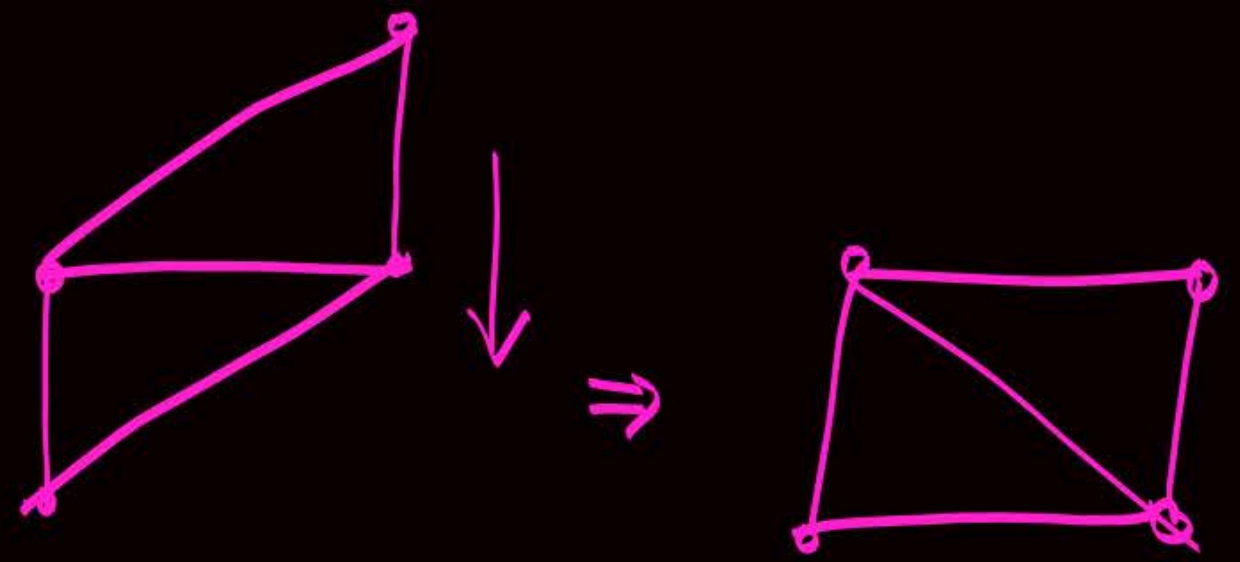
Q: Consider two graphs G_1 & G_2



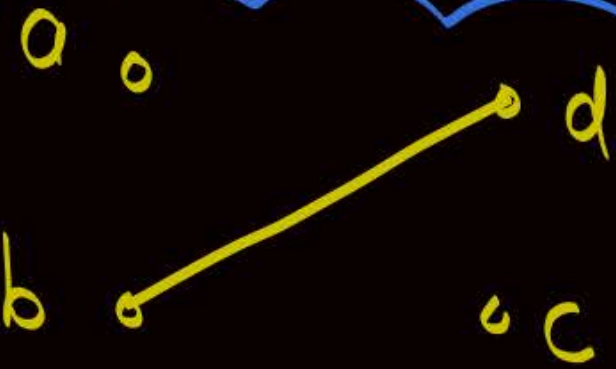
G_1



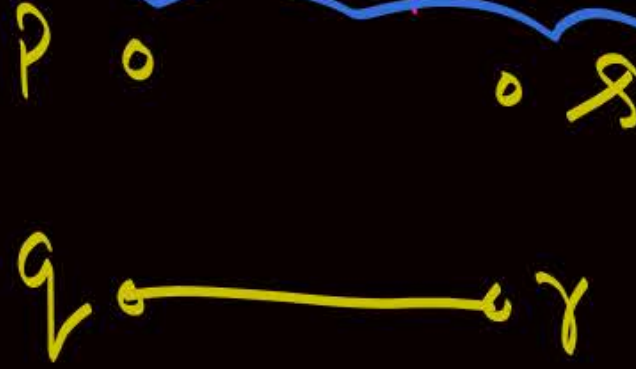
G_2



Check whether G_1 & G_2 are isomorphic or not?



$\overline{G_1}$



$\overline{G_2}$

We can observe that

$$\overline{G_1} \cong \overline{G_2}$$

$\therefore$ We can say that

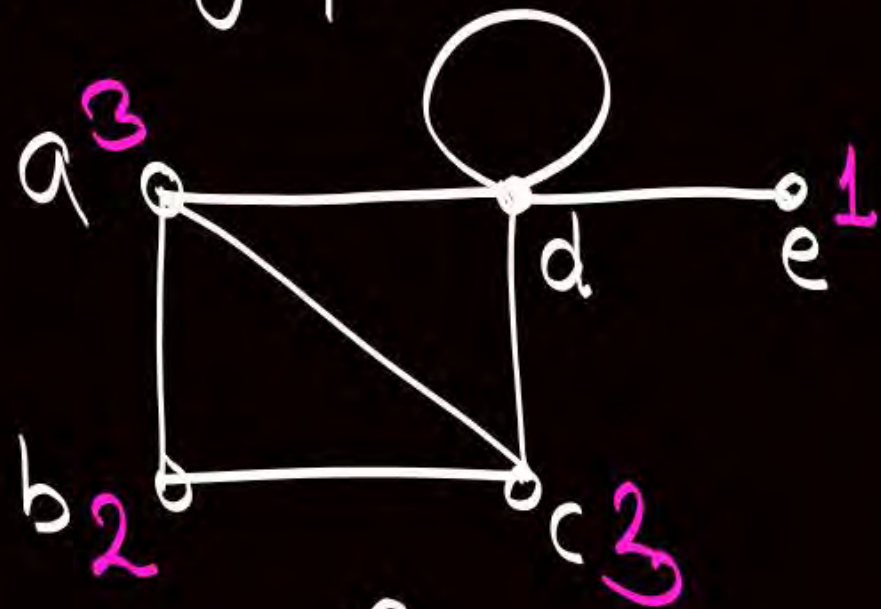
$$\boxed{G_1 \cong G_2} \quad \underline{\underline{\text{Ans}}}$$



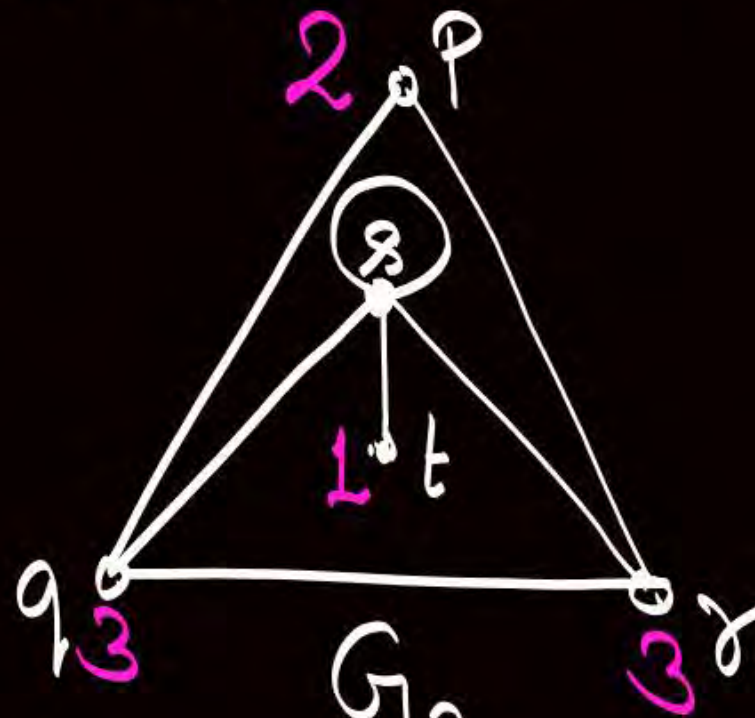
Topic : Graph Isomorphism

- Two graphs G_1 and G_2 are isomorphic if and only if their corresponding subgraphs {i.e the graph obtained by deleting some vertices from graph G_1 and the graph obtained by deleting the the images of those vertices from G_2 } are isomorphic to each other

Q: Consider two graphs G_1 and G_2 , and check whether the graphs are isomorphic or not?



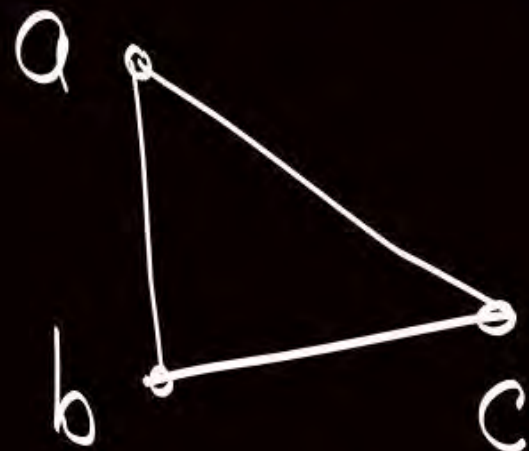
G_1



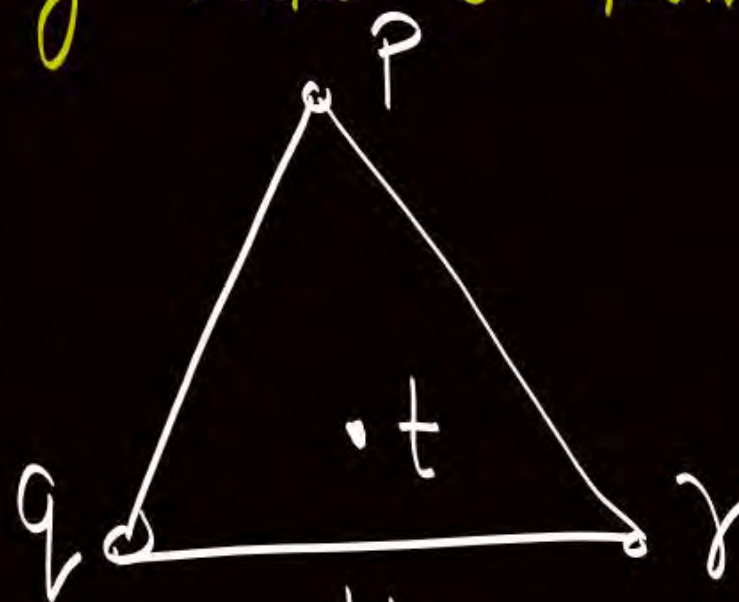
G_2

We can observe that image of vertex 'd' in G_1 is vertex 's' in G_2

Obtain the subgraph by deleting vertex 'd' from G_1 and its image 's' from G_2



H_1



H_2

We can observe

$$H_1 \cong H_2$$

$$\therefore \boxed{G_1 \cong G_2} \text{ Ans}$$

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '3' labeled vertices.

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '4' labeled vertices.

HW Q:- How many simple 'non-isomorphic' graphs are possible with n ($n \geq 4$) labeled vertices and two edges.

Today's Topic

Q: How many simple graphs are possible with '3' labeled vertices

Soln.

$${}^3C_2 = 2^3 = 8$$

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '3' labeled vertices.

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '3' labeled vertices.

With '3' vertices maximum number of edges possible in a simple graph are $= {}^3C_2 = 3$

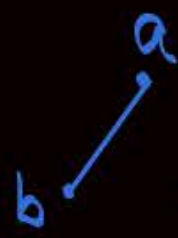
'Simple Graphs' possible with '3' vertices

0-edge:



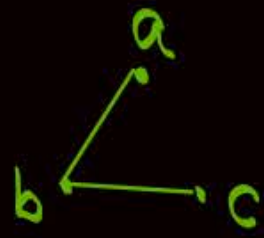
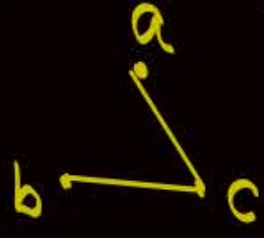
Only '1' graph with '3' vertices and 0-edge

'1' edge



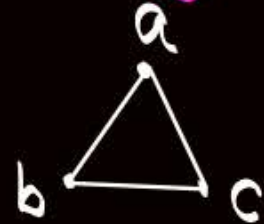
Three simple graphs are possible with '3' vertices and '1'-edges, but all of them are isomorphic to each other, so they will be counted only 'Once' in non-isomorphic graphs

'2' edge



Three simple graphs are possible with '3' vertices and '2'-edges, but all of them are isomorphic to each other, so they will be counted only 'Once' in non-isomorphic graphs

'3' edge



Only '1' graph with '3' vertices and 3-edges

Total no. of non-isomorphic graphs possible with 3 vertices

$$1 + 1 + 1 = 3$$

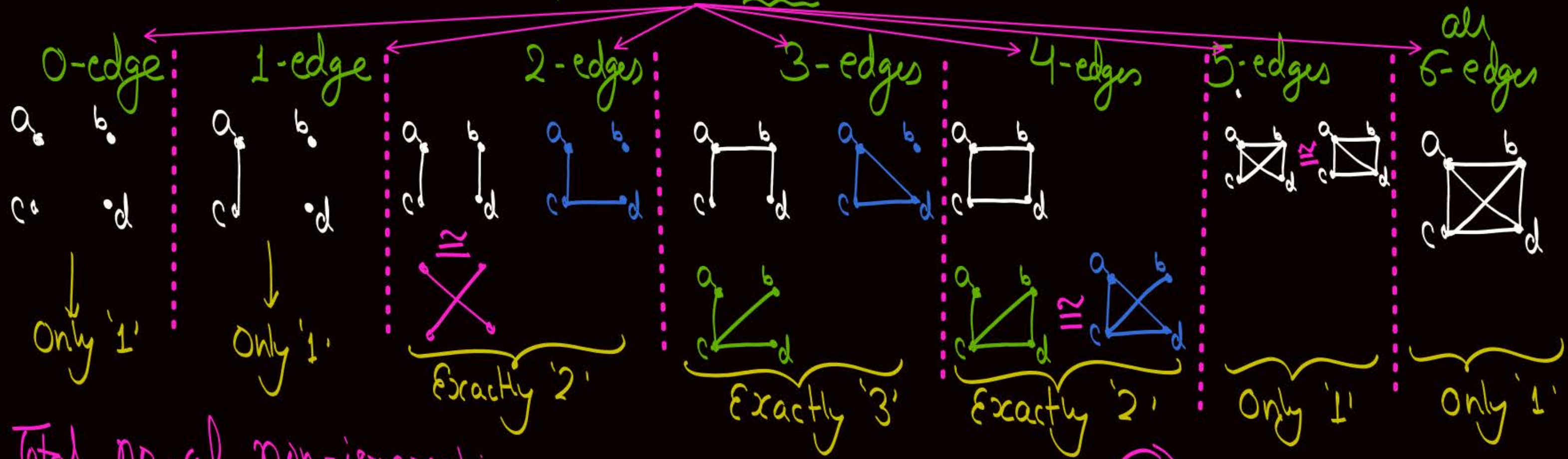
Ans = 4

Q: How many simple graphs are possible with '4' labeled vertices.

Solun = $(2)^{4C_2} = 2^6 = 64$

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '4' labeled vertices.

H.W. Q:- How many simple 'non-isomorphic' graphs are possible with '4' labeled vertices.
 With '4' vertices, maximum Number of edges possible are = ${}^4C_2 = 6$
 Simple 'non-isomorphic' graphs possible with '4' vertices and



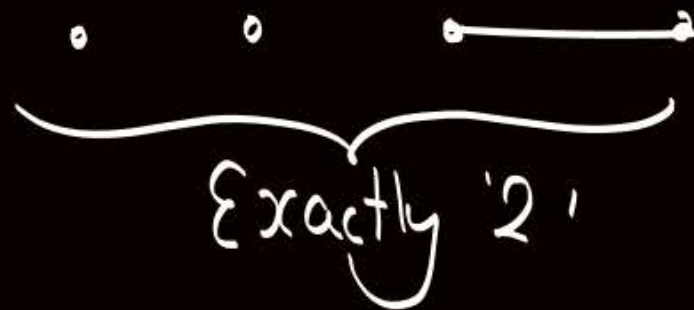
Total no. of non-isomorphic graphs possible with '4' vertices = $1 + 1 + 2 + 3 + 2 + 1 + 1 = 11$ Ans

Q:- How many different simple graphs are possible with upto '3' vertices.

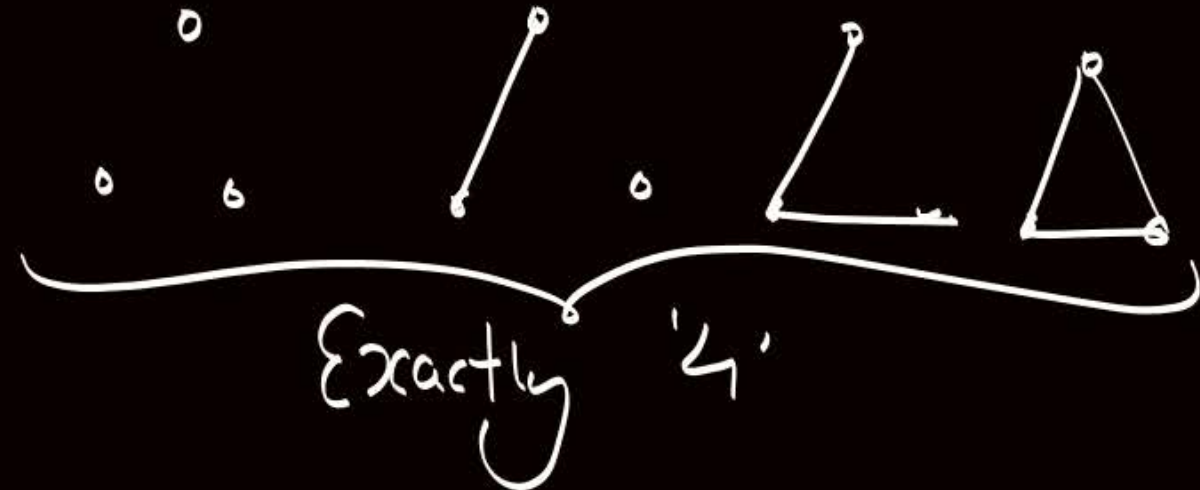
with '1' vertex



with '2' vertices



with '3' vertices



∴ Ans = $1 + 2 + 4 = \textcircled{7}$ Ans

Q:- How many simple 'non-isomorphic' graphs are possible with n ($n \geq 4$) labeled vertices and two edges.

Ans = 2

'1' when edges are not adjacent to each other

'1' when edges are adjacent to each other

When $n=1$: — No edge is possible

$n=2$: — No simple graph with '2' edges

$n=3$: — Only '1' non-isomorphic graph with '3' vertices and '2' edges

when $n \geq 4$

'1' : ^{w.r.t} adjacent edges

'1' : ^{w.r.t} non-adjacent edges

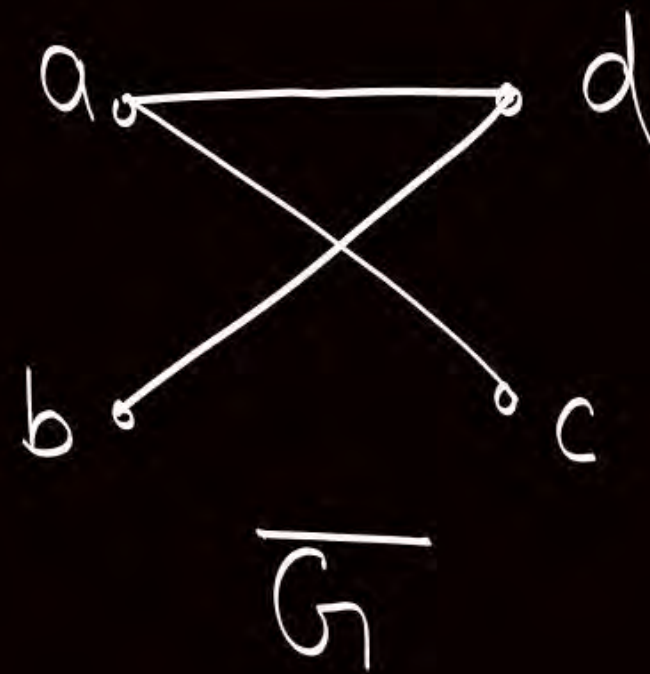
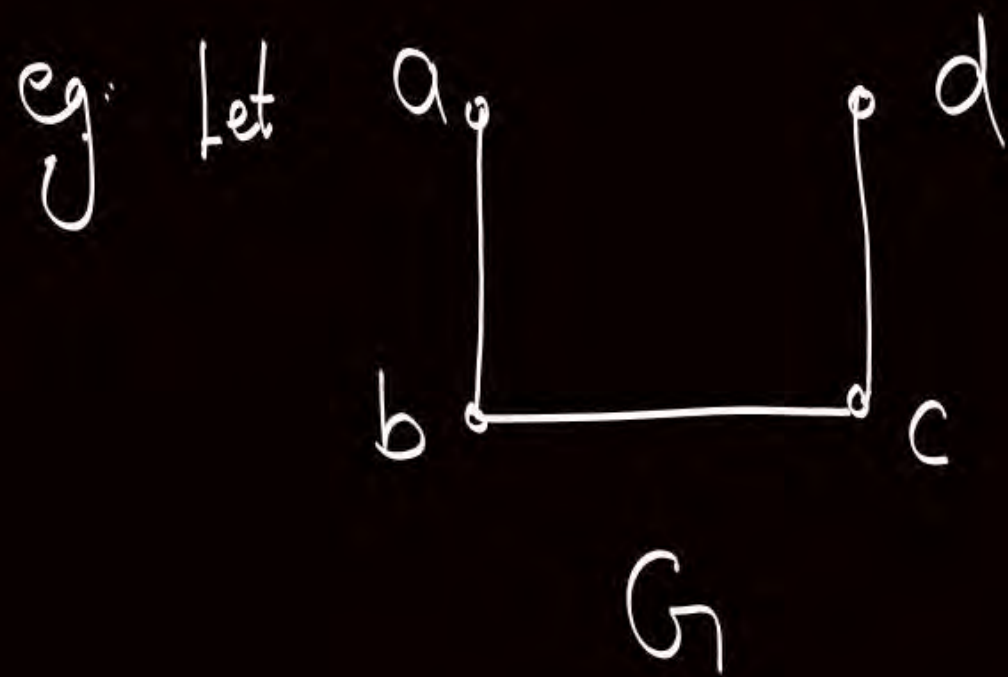


Topic : Self-Complementary Graph

- A simple graph G is said to be self-complementary if and only if complement of G is isomorphic to graph G itself.

ie, Graph G is a self complementary graph
if and only if

$$G \cong \overline{G}$$



We can observe $G \cong \overline{G}$

$\therefore$ Above graphs are self complementary graphs

Note

In a self Complementary graph G with 'n' vertices

i.e. When $G \cong \bar{G}$

① $|E(G)| = |E(\bar{G})|$ because of isomorphism — eqⁿ ①

② G & $\bar{G}$ are Complement of each other

∴ $|E(G)| + |E(\bar{G})| = \frac{n(n-1)}{2}$ — eqⁿ ②

By eqⁿ ① & eqⁿ ②

$$|E(G)| + |E(G)| = \frac{n(n-1)}{2}$$

$$|E(G)| = \frac{n(n-1)}{4}$$

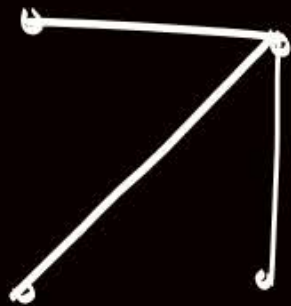
Note: If G is a self Complementary graph with ' n ' vertices
then $|E(G)| = \frac{n(n-1)}{4}$, but Converse of the statement
Need Not be true

eg. let $n=4$,
then $\frac{n(n-1)}{4} = \frac{4 \times 3}{4} = '3'$

let $G =$



$\bar{G} =$



Hence $E(G) = \frac{n(n-1)}{4}$, but ' G '
is not a self Complementary graph

It is a necessary
Condⁿ for graph to
be self Complementary,
but not sufficient

Note:

In a self Complementary graph ' G '

$$|E(G)| = \frac{n(n-1)}{4}$$

, and number of edges can never be in fraction.

∴ If graph G is a self Complementary graph then $\frac{n(n-1)}{4}$ must be an integer

∴ either ' n ' must be multiple of '4'

or
' $n-1$ ' must be multiple of 4

i.e, $n = 4K$ or $(4K+1)$

Where ' K ' may be any positive integer

If a graph is a self-Complementary graph, then $|V| = 4K$ or $4K+1$, but converse of the statement need not be true

In a Self Complementary graph G ,

$$|V(G)| \bmod 4 = 0 \text{ or } 1$$

Q:- Let ' C_n ' represent a cycle graph with ' n ' vertices.

If ' C_n ' is a self Complementary graph, then find the value of ' n '.

(i) C_n is a cycle graph
 $\therefore |E(C_n)| = n$ ——— eqⁿ ①

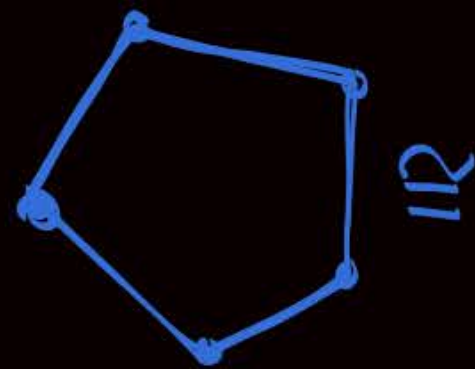
(ii) Given,
 C_n is a self Complementary graph
 $\therefore |E(C_n)| = \frac{n(n-1)}{4}$ ——— eqⁿ ②

By eqⁿ ① & eqⁿ ②

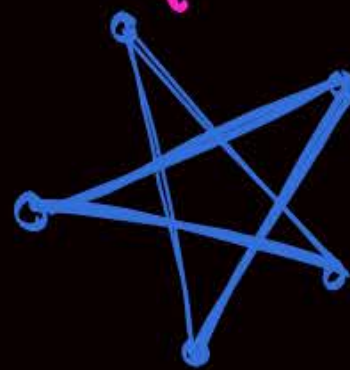
$$n = \frac{n(n-1)}{4}$$

$$4 = n-1$$

i.e. $n = 5$



or



it is also a cycle of length '5'



2 mins Summary



Topic

Graph isomorphism

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Self-complementary graph

Topic

Planar graphs

Topic

Region and degree of regions

Topic

Sum of degree theorem for regions

THANK - YOU